Actual transfers on the restriction's own field

8 October 2026 — analytical continuation and source-time decision

Conclusion. Specified one-interval fields close the traffic/receiving/ transfer/motion/count loop, including a local ramp extension. A nonuniform donor exposes two different temporal interpretations of sampled requests. No full multi-port numerical construction or useful-gradient result is claimed.

1. Common units, field and sampled requests

Distance is metres, time seconds, and population continuous vehicle-equivalent mass. Density ρ has units eq/m; flow F has units eq/s. Use

$$F(\rho) = \min\{u\rho, w(\kappa - \rho)\}, \qquad u = 30, \qquad w = 5, \qquad \kappa = 3/20, \tag{1}$$

$$\rho_c = \frac{w\kappa}{u+w} = 3/140, \qquad Q = u\rho_c = 9/14, \qquad L = 30, \qquad T = 1. \tag{2}$$

Each section obeys $\partial_t \rho + \partial_x F(\rho) = 0$ away from the fixed ports and moving restrictions. Source IT moves from (i, ℓ) to $(i+1, b)$ once. $\bar{\rho}_a = n_a/L$ is the donor section's average density; n_a is its mass. Native sampled sending/oriented requests are

$$S_a = T\min(u\bar{\rho}_a, Q) = \min(n_a, QT), \tag{3}$$

$$\theta_{ab} = \pi_{ab} T \ (b \neq a), \quad \theta_{aa} = 1 - \sum_{b \neq a} \theta_{ab}, \qquad d_{ab} = \theta_{ab} S_a. \tag{4}$$

d_{ab} is desired interval mass, whereas $r_{ab} = d_{ab}/T$ is its nominal rate. The source choice rate is $\pi_{ab} = (\bar{v}_b - \bar{v}_a)_+/(u\tau)$; $\bar{v}$ denotes the approved past-period lookahead Edie observation, not present point FD speed. Hold these choice observations and birth speeds at the public clock. Numerical internal events do not introduce new births or positive recovery. All records carry zero population.

For a fixed recipient port, its own right trace gives supply $R(t) = \min\{w(\kappa - \rho(t, 0+)), Q\}$. Donor own traces bound physically available sending. Old-record actual speeds use their own one-sided ahead traffic and the K.A running-minimum cap. In the examples below, that cap never falls below the held desired speeds. Both candidate source-time interpretations agree in these examples because donors stay available at the relevant rates.

Let $C(t) = n(0) + \int_0^t q_{\text{in}}(s)ds$ be the local count gauge and define

$$N(t, x) = C(t) - \int_0^x \rho(t, y)dy. \tag{5}$$

Then $N(t, L)$ is cumulative local exit, $N_x = -\rho$, $N_t = F$, and along an old record $z(t)$, $dN(t, z(t))/dt = F - \dot{z}\rho = 0$ under zero relative passing. These are the counts used below, not independently assigned labels.

2. Constrained through/lateral entry: exact whole interval

Initially the target section has ρ_c throughout. An old zero-passing record starts at $z(0) = 1$ with held desired speed $s = 5$. Own ahead traffic is free or vacuum, so $z(t) = 1 + 5t$. Its congested upstream state and flow are

$$\rho_U = \frac{w\kappa}{w+s} = 3/40, \qquad q_U = s\rho_U = 3/8. \tag{6}$$

The ordinary shock from ρ_c to ρ_U has speed

$$\sigma = \frac{q_U - Q}{\rho_U - \rho_c} = -5, \quad b(t) = 1 - 5t, \quad t_b = 1/5. \tag{7}$$

The downstream vacuum front is $e(t) = 1 + 30t$, reaching L at $t_e = 29/30$. For $0 < t < t_b$, the own field consists of ρ_c on $[0, b(t)]$, ρ_U on $[b(t), z(t)]$, vacuum on $[z(t), e(t)]$, and ρ_c from $e(t)$ to the exit when $e(t) < L$. For $t \geq t_b$, the queue starts at zero. Thus the derived boundary fluxes, not externally supplied histories, are

$$R(t) = q_{\text{in}}(t) = \begin{cases} 9/14, & 0 < t < 1/5, \\ 3/8, & 1/5 < t < 1, \end{cases} \quad q_{\text{out}}(t) = \begin{cases} 9/14, & 0 < t < 29/30, \\ 0, & 29/30 < t < 1. \end{cases} \tag{8}$$

The two sampled donor demands are through $r_1 = Q$ and lateral $r_2 = Q/3$. The lateral donor has choice fraction $1/3$ toward this target and $2/3$ toward its own next lane. For example $\tau = 2$, Edie target observation 30 and donor observation 10 give this fraction. The fast donor's sampled staying fraction is one. These are supplied history inputs; their reached-policy origin is not asserted. Both donors start congested, with own sending Q. IT gives

$$f_1(t) = \tfrac{3}{4}R(t), \quad f_2(t) = \tfrac{1}{4}R(t), \quad f_{2\to\text{own}} = \tfrac{2}{3}Q = 3/7. \tag{9}$$

The parallel recipient starts free at density $1/70$ with free exit, so its inflow and outflow are both $3/7$. No second advance is counted. Integrated accepted masses and the target state are

$$I = \int_0^1 R(t)dt = 3/7, \qquad I_1 = 9/28, \quad I_2 = 3/28, \quad O = 87/140, \tag{10}$$

$$n(1) = 9/14 + 3/7 - 87/140 = 9/20, \quad C(1) = 15/14, \tag{11}$$

$$z(1) = 6, \quad N(1,6) = 15/14 - 9/20 = 87/140 = N(0,1). \tag{12}$$

The ending field is $3/40$ on $[0, 6]$ and vacuum ahead. The native coarse step with initial receiving Q would transfer Q mass; the own causal field transfers $3/7$. This is the declared receiving adaptation being qualified, not a claim of identical coarse-step trajectories.

3. Donor fields and a fully available receiving case

To verify physical donor availability in section 2, start each 30 m donor at density 1/10, with inlet flow $1/4 = F(1/10)$. Their downstream flows are $f_1(t)$ and $f_2(t) + 3/7$. Every flow is below Q. The congested port density is $\rho^*(f) = \kappa - f/w$. On the congested branch all resulting waves travel upstream at -5. At $t = 1$ each donor has initial density on $[0, 25]$, first-period port density on $[25, 26]$, and final-period port density on $[26, 30]$:

Donor	First port density	Final port density	Ending mass
Through	3/56	3/32	41/14
Lateral + own through	9/280	51/1120	19/7

All port densities are at least ρ_c and below κ, hence sending stays Q. Wave positions stay away from the inlet. The masses also equal $3 + 1/4 - I_a$, with $I_a = 9/28$ and $15/28$. No unavailable mass is debited.

Available case. Use through/lateral rates 3/20 and 1/10; total $q_0 = 1/4$. The target starts uniformly at $\rho_0 = q_0/u = 1/120$, with the same old record at 1 m and desired speed 5. Its queue rear moves at

$$\sigma_0 = \frac{3/8 - 1/4}{3/40 - 1/120} = 15/8 > 0. \tag{13}$$

Consequently the inlet stays free: $R(t) = Q$, $q_{\text{in}} = 1/4$, both requests are fully served. At the end its field is 1/120 on $[0, 23/8]$, 3/40 on $[23/8, 6]$, and vacuum ahead; the original downstream free tail has exited. Direct integration and conservation agree:

$$I = 1/4, \quad O = (1/4)(29/30) = 29/120, \tag{14}$$

$$n(1) = (1/120)(23/8) + (3/40)(25/8) = 31/120, \tag{15}$$

$$C(1) = 1/2, \qquad N(1, 6) = 29/120 = N(0, 1). \tag{16}$$

The through donor is uniformly free at density 1/200 and supplies 3/20. The lateral donor has saturated demand and choice fraction 7/45, yielding lateral 1/10 and own through 19/35. With the same $\tau = 2$, target Edie observation 30 and donor observation 62/3 produce 7/45. Its total outflow is Q; a congested initial field can sustain that with upstream waves. Its parallel recipient starts free at $(19/35)/30 = 19/1050$ and receives/exits 19/35. Both cases honor the separate past-speed convention.

4. Local ramp, multiple records and sampled births

Return to section 2's target. Add a saturated finite ramp. Let its fixed priority be illustratively $p = 1/3$, unchanged within the calculation. Non-ramp total desired rate is $4Q/3$; it includes both through and lateral traffic. Ramp demand is Q. The approved shared node gives

$$f_r = R/3, \quad f_m = 2R/3, \quad f_{\text{through}} = R/2, \quad f_{\text{lateral}} = R/6. \tag{17}$$

All three still total R; receiving is spent once. Integrated masses are $1/7, 3/14, 1/14$. With 30 m congested donors initially at $1/10$ and inlets $1/4$, all resulting port states are congested and admissible. Their ending masses are, respectively, $87/28, 85/28, 11/4$; the lateral donor's $11/4$ also debits its parallel through transfer $3/7$. The ramp's upstream receiving stays $1/4$ because its downstream waves cannot reach the upstream boundary in one second. M.A may supply $1/4$ there under adequate offered flow and permission. This is compatible upstream admission, not a new contemporaneous cap on actual merge entry.

A second old record. Put another zero-passing record at 2 m, with held desired speed 15. Its own upstream state is $\rho_{U2} = 3/80$. Initially its queue rear is $2 - 5t$, front $z_2(t) = 2 + 15t$, downstream vacuum front $2 + 30t$. The first record's vacuum front meets this queue rear at

$$1 + 30t = 2 - 5t \quad \Longrightarrow \quad t_* = 1/35. \tag{18}$$

At that contact, the ordinary vacuum-to-ρ_{U2} shock has speed 15; its continuing rear is $b_2(t) = 10/7 + 15t$. Its queue therefore has constant length $4/7$ and mass $3/140$ after t_*. No passing or coincident-record jitter is introduced. The first record stays at speed 5; any encountered second queue moves at 15 and does not require slower braking. The second's ahead traffic is free/vacuum, so its own cap permits 15. The inlet wave in section 2 is unaffected. Exit flow is Q until $14/15$, then zero. At the end the target field is

$$\rho(1,x) = \begin{cases} 3/40, & 0 < x < 6, \\ 3/80, & 115/7 < x < 17, \\ 0, & \text{elsewhere in } (0,30), \end{cases} \qquad n(1) = 9/20 + 3/140 = 33/70. \tag{19}$$

Conservation gives $9/14 + 3/7 - 3/5 = 33/70$. The count labels are

$$C(1) = 15/14, \quad h_1 = N(1,6) = 87/140 = N(0,1), \quad h_2 = N(1,17) = 3/5 = N(0,2). \tag{20}$$

All queue and gap masses agree with these counts.

Completed E1 attachment. For illustrative threshold one equivalent, past ramp/lateral counters $9/10$ and $19/20$ become $73/70$ and $143/140$. Each crosses one threshold; remainders $3/70$ and $3/140$ stay recorded. At $t = 1$ insert two distinct records at the recipient inlet, both with count $N(1,0) = 15/14$, retaining request-phase desired speeds (e.g. 5 and 10). They add no mass and do not change preceding receiving. The own queue ahead has speed 5, so its trace supplies the immediate next-interval cap. Neither arbitrary later contact continuation nor section crossings are proved here.

5. Nonuniform sending: a causal bound and two distinct completions

Take a finite ramp approach of length 30 with initial $\rho = 3/20$ on $[0, 3]$ and vacuum on $(3, 30)$. Close upstream admission this interval. Give its merge ample receiving, with no competing mainline demand; N.A then reallocates the unused mainline share. Initial mass is $n_0 = 9/20$ and the source average gives

$$S = T \min(un_0/L, Q) = 9/20. \tag{21}$$

The ordinary jam-to-vacuum triangular fan has edges $3 - 5t$ and $3 + 30t$, with critical density between. The first traffic reaches the exit at

$$t_a = (30 - 3)/30 = 9/10, \qquad \int_0^1 f(t)dt \le Q(1 - t_a) = 9/140 < 9/20. \tag{22}$$

This bound follows from the own field; an interpolation cannot physically deliver the full coarse accepted S in this interval. CTM's original uniform internal approximation is not refuted by this comparison.

T.A: interval mass ceiling. Let actual flow be the available own flow until total request S is spent, also respecting receiving. Here $f_A(t) = 0$ before t_a and Q after; the ceiling never binds. Thus

$$I_A = 9/140, \quad n_A(1) = 9/20 - 9/140 = 27/70. \tag{23}$$

With the closed inlet, the rear fan reaches $x = 0$ at 3/5 s. The ensuing vacuum-to-critical rear moves at u, reaching $x = 12$ at $t = 1$. The ending field is ρ_c on $[12, 30]$, vacuum before. Its mass is $(18)(3/140) = 27/70$.

T.B: held nominal request rate. Impose additionally $f_B \le S/T$, while never exceeding own availability. Here $f_B = 0$ until t_a, then 9/20 instead of Q. Its own outlet queue state is

$$\rho_B^* = \kappa - (S/T)/w = 3/50, \quad I_B = (9/20)(1/10) = 9/200, \tag{24}$$

$$n_B(1) = 9/20 - 9/200 = 81/200. \tag{25}$$

The queue rear travels at -5 from the exit after t_a; it reaches $x = 59/2$ at $t = 1$. The ending field is ρ_c on $[12, 59/2]$ and 3/50 on $[59/2, 30]$, with vacuum before. Its mass is exactly 81/200. Both completions are conservative, causal and jam-bounded; the second has an additional nominal-rate constraint. They differ by 27/1400 mass. An initially empty 30 m recipient with free exit contains the transferred traffic on $[0, 3]$ at the ending clock: density 3/140 under T.A or 3/200 under T.B. Its mass is respectively I_A or I_B, with no exit yet. Thus these are paired completed merge entries, not merely donor debits.

For an eligible actual ramp-entry counter initially $H_0 = 47/50$, E1 yields

$$H_A(1) = 703/700 > 1, \qquad H_B(1) = 197/200 < 1. \tag{26}$$

T.A creates one next-clock effect (remainder 3/700), whereas T.B creates none. This is an operator diagnostic of the approved birth threshold, not an optimized-control witness or a proposal to manufacture hidden information.

6. Why the mass ceiling is natural, and what it does not settle

For a section with $L = uT$, all upstream-port admissions strictly after the current clock need at least T to traverse its full length because actual forward speeds do not exceed u. They cannot leave this section within the same regular interval. Lane transfers also enter its upstream port. With conservation and ordinary stationary-port capacity Q,

$$\text{actual total outgoing mass} \leq \min\{n(0), QT\} = S. \tag{27}$$

This excludes interior sources and zero-length shortcuts, neither of which is approved here. It supports T.A's total sending ceiling without requiring a second held rate. It does not assign exact movement-specific allowances under selective target blockage: $d_{ab} = \theta_{ab} S_a$ still needs explicit unspent-claim accounting, donor joint availability and target competition. A naive repeated use of instantaneous choice fractions may retry the same traffic and exceed a movement's sampled desired mass.

For T.A, proposed numerical claim state is

$$b_{ab}(t) = d_{ab} - \int_0^t f_{ab}(s)ds \geq 0. \tag{28}$$

It is not extra population or a vehicle identity. The common constraints include $\sum_b f_{ab} \leq D_a^{\text{own}}$, target shared receiving, and zero flow when its claim is exhausted. N.A and IT competition must have a declared meaning while positive claims are temporarily unavailable or become depleted. Those constraints alone do not define the full multi-port allocation. T.B instead holds $r_{ab} = d_{ab}/T$ as nominal rates, additionally reducing them for unavailable donor traffic or constrained receiving; it likewise needs a qualified shared-port transcription. No new upstream control is proposed.

The methods differ at finite public resolution. Their relation to the source's approximation/refinement and their coupled motion/birth histories must be qualified, not asserted from dimensional conversion. Because they change those histories, Q.J.F.T is a shared methodological choice rather than a private solver setting. T.A is provisionally preferred, not adopted.

7. Primary basis and evidence limits

Laval and Daganzo (2006), *Lane-changing in traffic streams*, sections 2.1–2.2, equations 5–13 and footnote 2, supplies desired sampled advancing masses, Edie observations and IT. Its original numerical/source recipe is not claimed to prove the present adaptation.

Jin, arXiv:0810.3952v2, sections 3–5, distinguishes discrete local merge fluxes and continuous Riemann solutions. Interior states can represent the limit; section 4.1 proves the fixed-priority flux's invariance. A simple two-incoming-road theorem does not establish the present oriented multilanes. No new priority is inferred for IT from that theorem.

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The examples use admissible specified initial/history state; no reached public scenario, arbitrary contact/crossing solver, clearance, full TST or useful gradient is inferred. The four-step pass stops at Q.J.F.T. A complete qualified finite construction can still meet Q.J.F without a universal continuum existence theorem. Existing M.A/N.A/G1/E1/K.A remain approved; representation A1/A2/C and the complete law remain unadopted. Stage 4 unopened.